Prayas JEE 3.0 2025

Mathematics

Trigonometric Equation

DPP: 3

Q1 General solution of the equation $\cot\theta - \tan\theta = 2$ is
(A) $\mathrm{n}\pi + \frac{\pi}{4}$
(B) $\frac{n\pi}{2} + \frac{\pi}{8}$
(C) $\frac{\mathrm{n}\pi}{2} \pm \frac{\pi}{8}$
(D) None of these

Q2 If $\sqrt{3}\cos\theta + \sin\theta = \sqrt{2}$, then the most general value of θ is
(A) $\mathrm{n}\pi + (-1)^{\mathrm{n}}\frac{\pi}{4}$
(B) $(-1)^{\mathrm{n}}\frac{\pi}{4} - \frac{\pi}{3}$
(C) $n\pi + \frac{\pi}{4} - \frac{\pi}{3}$
(D) $\mathrm{n}\pi + (-1)^{\mathrm{n}}\frac{\pi}{4} - \frac{\pi}{3}$

Q3 The number of the solutions of the equation $3\sin x + 4\cos x - x^2 - 16 = 0$ is
(A) 3 (B) 2
(C) 1 (D) 0

Q4 The number of solution of equation $\mathrm{x}^3 + \mathrm{x}^2 + 4x + 2\sin x = 0$ in $0 \leq x \leq 2\pi$ is
(A) Zero (B) One
(C) Two (D) Four

Q5 If $\sin x + \cos x = \sqrt{y + \frac{1}{y}}, x \in (0, \pi)$, then
(A) $\mathrm{x} = \frac{\pi}{4}, \mathrm{y} = 1$
(B) $y = 0$
(C) $y = 2$
(D) $x = \frac{3\pi}{4}$

Q6 The number of solution of the equation $1 + \sin x \sin^2\frac{x}{2} = 0$ in $[-\pi, \pi]$, is
(A) 0 (B) 1
(C) 3 (D) none of these

Q7 One root of the equation $\cos x - x + \frac{1}{2} = 0$ lies in the interval
(A) $\left[0, \frac{\pi}{2}\right]$
(B) $\left[-\frac{\pi}{2}, 0\right]$
(C) $\left[\frac{\pi}{2}, \pi\right]$
(D) $\left[\pi, \frac{3\pi}{2}\right]$

Q8 If $\mathrm{a}, \mathrm{b} \in [0, 2\pi]$ and equation $\mathrm{x}^2 - 2\mathrm{x} + 4 = 3\sin(\mathrm{ax} + \mathrm{b})$ has at least one solution and the least positive value of $\mathrm{a} + \mathrm{b}$ is k, then $\frac{2\mathrm{k}}{\pi}$ is equal
(A) 0 (B) 1
(C) 2 (D) 3

Q9 The number of real solutions of the equation $(\sin x - x)\left(\cos x - x^2\right) = 0$ is
(A) 1 (B) 2
(C) 3 (D) 4

Q10 If $0 < \theta < 2\pi$, then the intervals of values of θ for which $2\sin^2\theta - 5\sin\theta + 2 > 0$ is
(A) $\left(0, \frac{\pi}{6}\right) \cup \left(\frac{5\pi}{6}, 2\pi\right)$
(B) $\left(\frac{\pi}{8}, \frac{5\pi}{6}\right)$
(C) $\left(0, \frac{\pi}{8}\right) \cup \left(\frac{\pi}{6}, \frac{5\pi}{6}\right)$
(D) $\left[0, \frac{\pi}{6}\right) \cup \left(\frac{5\pi}{6}, 2\pi\right]$

Answer Key

Q1	(B)	Q6	(A)
Q2	(D)	Q7	(A)
Q3	(D)	Q8	(B)
Q4	(B)	Q9	(C)
Q5	(A)	Q10	(A)